Supplementary Material for "Leakage-Aware, Sensitivity-First Validation of Blood-Smear Leukemia Classifiers: A Two-Dataset Study of Engineered Features, Deep Features and Dimensionality Reduction"

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This document contains the complete settings, computing environment, full parameter sweeps and secondary analyses referred to in the main text. Code, split files and per-model outputs are released with the paper.

S1. Settings and computing environment
S2. Evaluation protocols and segmentation

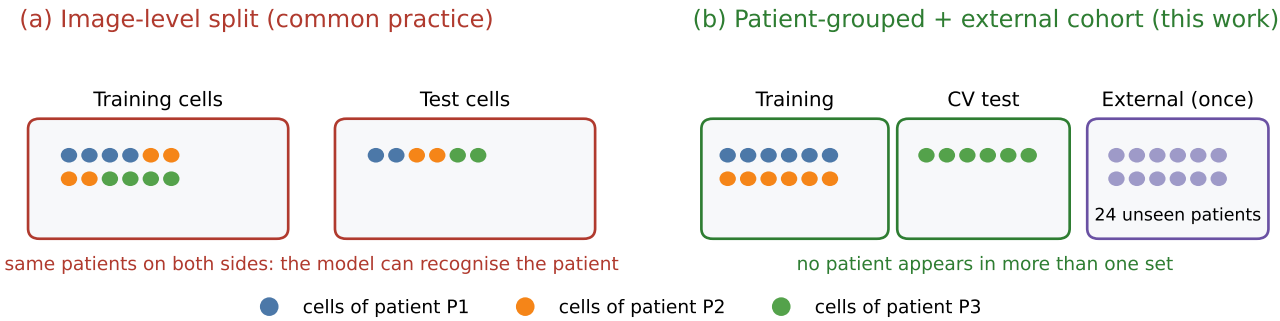


Fig. S1. Evaluation protocols. (a) Image-level splitting places cells of the same patient in training and test sets. (b) Patient-grouped splitting with an external cohort, used in this work; the external cohort is scored once.

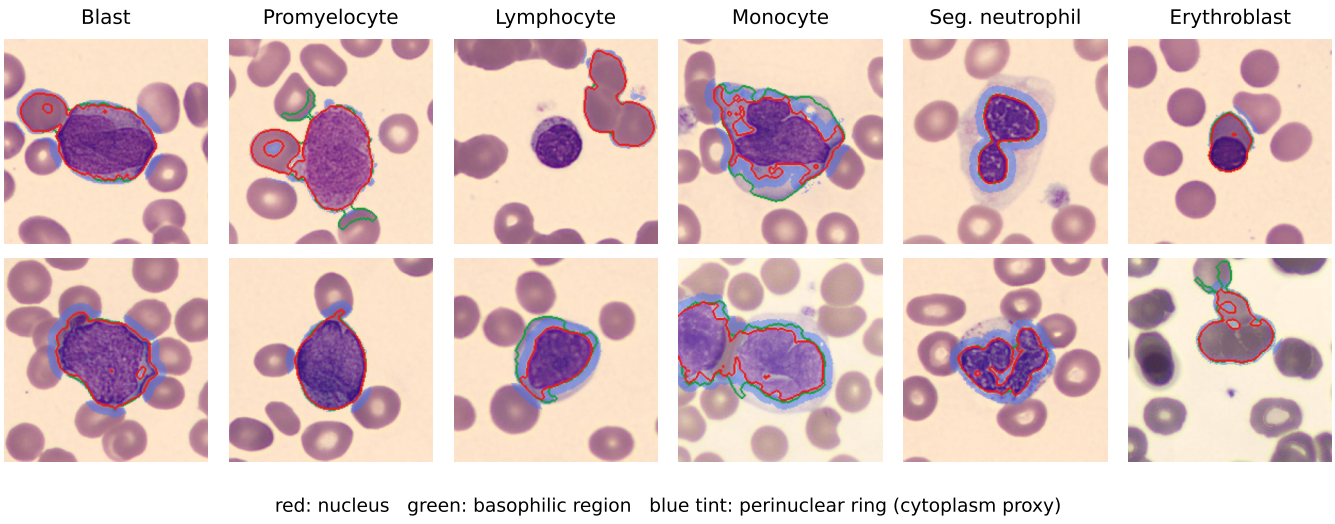


Fig. S2. Segmentation behind the engineered features, shown on cells at the 25th and 75th percentile of each class list. Red: nucleus; green: basophilic region; blue tint: perinuclear ring. The upper lymphocyte and lower erythroblast illustrate failures in which the mask falls on neighboring red cells.

S3. Feature engineering and full PCA sweep

S4. Training settings and learning curves

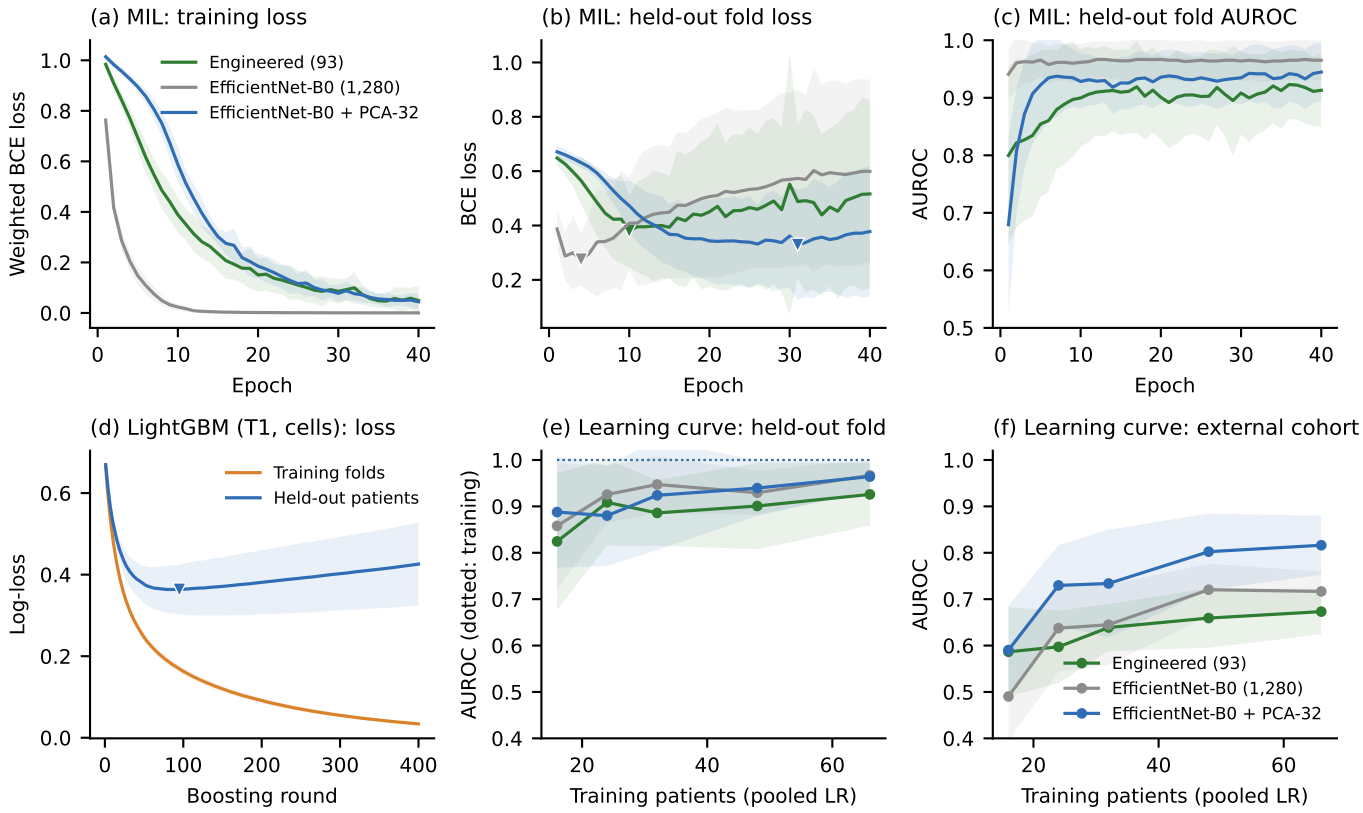


Fig. S3. Training dynamics on the Discovery cohort (mean ± SD over 5 folds × 3 seeds). (a)–(c) Gated attention MIL trained with AdamW at a constant learning rate of 2×10^{-4}: training loss (class-weighted binary cross-entropy), loss and AUROC on the held-out fold per epoch; triangles mark the epoch with the lowest held-out loss. (d) Cell-level LightGBM for T1 (learning rate 0.05): log-loss on training folds and on held-out patients per boosting round. (e), (f) Learning curves of the pooled patient model versus the number of training patients, on the held-out fold (dotted: training AUROC) and on the external cohort. Values in (e) and (f) are for single fold models, not the fold ensemble used in the tables, and are therefore slightly lower.

S5. Patient-level models without feature engineering

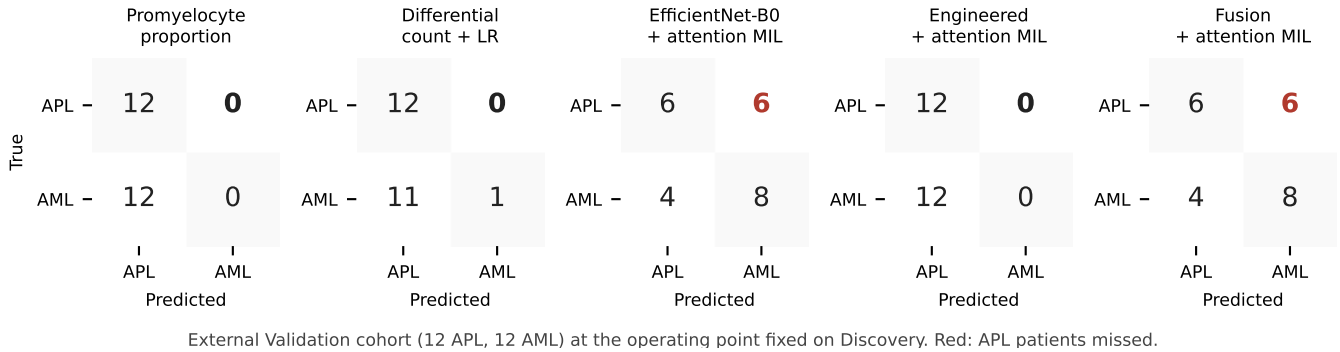


Fig. S4. Confusion matrices on the external cohort at the operating point fixed on Discovery for a sensitivity of at least 0.95. Missed APL patients are shown in red.

S6. Pooled 106-patient analysis and local recalibration

S7. Ablation of engineered feature groups

References

[1] F. Pedregosa, G. Varoquaux, A. Gramfort *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

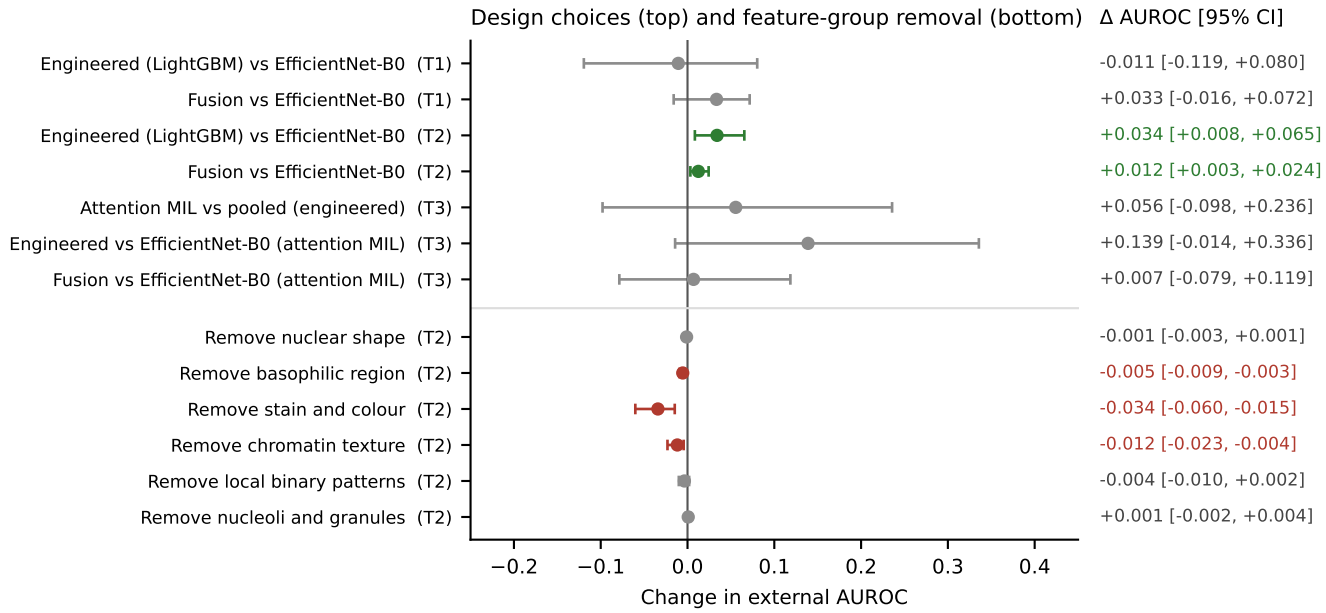


Fig. S5. Effect of design choices (top) and of removing each engineered feature group for blast detection (bottom) on external AUROC. Bars are 95% confidence intervals from a paired patient-level bootstrap.

TABLE S1
COMPLETE SETTINGS OF EVERY COMPONENT. ALL VALUES WERE FIXED BEFORE THE VALIDATION COHORT WAS SCORED; NONE WAS TUNED ON IT. DEFAULTS OF THE NAMED LIBRARIES APPLY TO ANYTHING NOT LISTED.

Component	Setting	Value
Data and splits	Development / external cohorts	APL-AML: Discovery (82 patients) / Validation (24 patients), split by diagnosis date by the data owners
	Second dataset	C-NMC 2019: official folds 0–1 (42 subjects) / fold 2 (31 subjects), designated before any analysis
	Cross-validation	5 folds; image-level `StratifiedKFold` or patient-level `StratifiedGroupKFold` (cells), `StratifiedKFold` over patients (T3)
	Random seeds	0, 1, 2 (fold assignment, cell sampling, model initialization); results are averaged over seeds
	Cells per patient (T3)	Up to 500, sampled at random once (seed 0); patients with fewer cells keep all
Engineered features (93)	Input	RGB, bilinear resize to 192×192
	Color deconvolution	Ruifrok–Johnston H&E matrix (`skimage rgb2hed`)
	Nucleus	Gaussian smoothing ($\sigma = 1$) of hematoxylin, Otsu threshold, opening (disk 2), component nearest the center (within $0.30\times$ image size); fallback: darkest 15% of gray levels
	Cytoplasm proxies	Basophilic region: blue-dominant pixels within one nuclear radius, opening (disk 2), closing (disk 4); ring: 2–9 px from the nucleus, excluding pixels brighter than the 90th lightness percentile -3
	Texture	GLCM on 33 gray levels, distances 1, 2, 4 px, angles $0, \pi/4, \pi/2, 3\pi/4$ (contrast, dissimilarity, homogeneity, energy, correlation; averaged over angles); uniform LBP with $(P, R) = (8, 1)$ and $(16, 2)$
	Missing values	Unreadable images give NaN rows, imputed with the training-fold median
	C-NMC background	Pixels with brightest channel < 15 set to white before feature extraction (engineered features only)
Deep features	Encoder	EfficientNet-B0, ImageNet weights (`timm`), frozen, global-average-pooled 1,280-d output
	Input	Bicubic resize to 224×224, ImageNet mean/std normalization, no augmentation
	Extraction	Batch size 64, 2 data-loader workers, CPU, `torch.no_grad`
Feature engineering	PCA	Standardization then PCA (`svd_solver`=auto, `random_state`=0), both fitted on training-fold cells only; 16, 32, 64 components pre-specified; 4, 8, 128, 256 as sensitivity analysis; pooled component statistics divided by one common scale (no whitening); per-component standardization reported for comparison
	Correlation filter	Drop the later feature of every pair with $\lvert r\rvert > 0.9$ (training cells, $\leq$20,000 sampled)
	Stain normalization / L1	Per-slide median/SD standardization of the 26 stain features; L1-penalized LR, $C = 0.5$, SAGA solver
Cell-level classifiers (T1, T2)	Logistic regression	L2, $C = 0.1$, L-BFGS, 3,000 iterations, class-balanced weights; median imputation and standardization inside the pipeline
	LightGBM	400 trees, learning rate 0.05, 31 leaves, row subsampling 0.8 (every iteration), column subsampling 0.8, class-balanced weights, 2 threads
	Calibration	Temperature scaling fitted on an inner 80/20 split of the training fold (bounded search 0.05–20)
Pooled patient model (T3)	Patient vector	Mean, SD, 10th, 50th, 90th percentile of every cell feature ($5\times$ dimensions)
	Classifier	LR, L2, $C = 0.1$, 5,000 iterations, class-balanced weights, standardization inside the pipeline
Gated attention MIL (T3)	Architecture	Linear($d \to 128$)–ReLU–Dropout(0.25); gated attention with $\tanh$ and sigmoid branches of 64 units; linear classifier on the attention-weighted mean
	Optimization	AdamW, learning rate 2×10^{-4} (constant, no schedule), weight decay 10^{-3}, one bag per step, 40 epochs, no early stopping
	Loss	Binary cross-entropy with positive-class weight $n_{\mathrm{AML}}/n_{\mathrm{APL}}$ of the training fold
	Augmentation	Random removal of 20% of a bag's cells at every step (bags with more than 10 cells)
	Input scaling	Median imputation and standardization from training-fold cells
Fine-tuned CNN (T1, T2)	Trainable part	EfficientNet-B0 stages 6–7, 1×1 head convolution (1,280 channels), new linear classifier (dropout 0.2); stem and stages 1–5 frozen, their $112 \times 14 \times 14$ outputs cached once (float16)
	Optimization	AdamW, peak learning rate 3×10^{-4}, 1 warm-up epoch then cosine decay, weight decay 10^{-4}, batch 128, 6 epochs, class-weighted binary cross-entropy
	Data	Up to 5,000 training cells per fold (random); 15% of training patients (grouped split) or images (image split) as inner validation for early stopping by validation loss; one seed
	Augmentation	Random horizontal and vertical flips and 90° rotations of the cached

TABLE S2
COMPUTING ENVIRONMENT AND RUN TIMES.

Item	Specification
Hardware	Intel Xeon 2.80 GHz, 2 vCPUs, 7.8 GB RAM, no GPU
Software	Ubuntu 24.04.5 LTS; Python 3.13.16; PyTorch 2.14.1 (CPU); scikit-learn 1.9.1 [1]; LightGBM 4.7.0; scikit-image 0.26.0; timm 1.0.30
Random seeds	0, 1, 2
Engineered features	16.1 min for 25,724 cells (2 processes)
EfficientNet-B0 embeddings	51.2 min for 25,724 cells (2 threads)
Cell-level model, 5-fold CV	11–40 s (LR), 15–600 s (LightGBM)
Attention MIL, 5-fold CV	33–61 s per seed

TABLE S3
FEATURE ENGINEERING FOR PATIENT-LEVEL APL DETECTION. ALL STEPS ARE FITTED INSIDE THE DISCOVERY TRAINING FOLDS; THE VALIDATION COHORT IS SCORED ONCE. Δ: EXTERNAL AUROC CHANGE RELATIVE TO THE UNREDUCED REPRESENTATION (PAIRED PATIENT BOOTSTRAP). THE CONFIGURATION WITH THE HIGHEST DISCOVERY CV AUROC (PRE-DECLARED SELECTION RULE) IS MARKED *. ORACLE COLUMN: SPECIFICITY AT 100% RECALL IF THE THRESHOLD WERE RE-SET ON THE VALIDATION COHORT (REFERENCE ONLY, NOT DEPLOYABLE).

Representation	Engineering step	CV AUROC	External AUROC [95% CI]	Δ [95% CI]	Recall (missed)	Spec.	F1	Oracle spec.
Engineered, all 93	–	0.926	0.729 [0.511, 0.924]	–	0.58 (5/12)	0.67	0.61	0.33
Engineered	Correlation filter ($\lvert r\rvert > 0.9$)	0.922	0.736 [0.529, 0.930]	+0.007 [-0.070, +0.084]	0.58 (5/12)	0.67	0.61	0.42
Engineered	Per-slide stain normalization	0.872	0.576 [0.329, 0.818]	-0.153 [-0.326, +0.021]	0.67 (4/12)	0.50	0.62	0.08
Engineered	L1 (LASSO) selection	0.898	0.590 [0.350, 0.820]	-0.139 [-0.274, -0.023]*	0.75 (3/12)	0.42	0.64	0.17
Engineered	Filter + stain normalization	0.869	0.569 [0.322, 0.798]	-0.160 [-0.333, +0.009]	0.67 (4/12)	0.50	0.62	0.08
EfficientNet-B0, 1,280-d	–	0.957	0.743 [0.538, 0.926]	–	0.50 (6/12)	0.67	0.55	0.42
EfficientNet-B0	PCA, 16 components	0.909	0.792 [0.580, 0.963]	+0.049 [-0.096, +0.213]	0.92 (1/12)	0.67	0.81	0.42
EfficientNet-B0	PCA, 32 components	0.958	0.833 [0.644, 0.972]	+0.090 [-0.014, +0.210]	0.50 (6/12)	0.92	0.63	0.42
EfficientNet-B0*	PCA, 64 components	0.966	0.819 [0.604, 1.000]	+0.076 [-0.171, +0.324]	0.50 (6/12)	0.92	0.63	0.00
Fusion	Filter + PCA-32	0.960	0.764 [0.566, 0.938]	+0.021 [-0.105, +0.140]	0.58 (5/12)	0.83	0.67	0.42
Digital differential	8 predicted cell classes	0.852	0.785 [0.570, 0.953]	+0.056 [-0.104, +0.229]	1.00 (0/12)	0.33	0.75	0.50
Differential + engineered	Filtered	0.928	0.764 [0.562, 0.943]	+0.035 [-0.015, +0.111]	0.58 (5/12)	0.67	0.61	0.50
Engineered, attention MIL	Filter + stain normalization	0.922	0.569 [0.326, 0.793]	-0.160 [-0.422, +0.091]	0.42 (7/12)	0.75	0.50	0.17
EfficientNet-B0, attention MIL	PCA-32	0.924	0.799 [0.580, 0.972]	+0.069 [-0.133, +0.281]	0.50 (6/12)	0.75	0.57	0.50

Recall, specificity and F1 at the operating point fixed on Discovery (target sensitivity 0.95). * $p < 0.05$ unadjusted; † $p < 0.05$ after Holm correction across all comparisons in the table.

TABLE S4
PATIENT/SUBJECT-LEVEL DETECTION BEFORE AND AFTER PCA OF THE EFFICIENTNET-B0 EMBEDDINGS (POOLED-STATISTICS LR), ON BOTH DATASETS. PCA IS FITTED ON TRAINING-FOLD CELLS ONLY; THE PRINCIPAL-COMPONENT SCORES KEEP THEIR RELATIVE VARIANCES (NO WHITENING). STD.: EXTERNAL AUROC WHEN EACH COMPONENT IS INSTEAD STANDARDIZED TO UNIT VARIANCE BEFORE THE CLASSIFIER (WHITENING), AS IN OUR FIRST ANALYSIS. Δ: CHANGE IN EXTERNAL AUROC RELATIVE TO NO PCA UNDER THE SAME SCALING (PAIRED BOOTSTRAP). RECALL, SPECIFICITY AT THE OPERATING POINT FIXED ON DEVELOPMENT DATA. COMPONENTS 16, 32 AND 64 WERE PRE-SPECIFIED.

	APL-AML (24 external patients, 12 APL)						C-NMC (31 external subjects, 17 ALL)					
Components	Var.	Std.	External [95% CI]	Δ	Recall (missed)	Spec.	Var.	Std.	External [95% CI]	Δ	Recall (missed)	Spec.
None (1,280)	1.00	0.743	0.708 [0.493, 0.907]	–	0.67 (4/12)	0.67	1.00	0.908	0.908 [0.786, 0.992]	–	1.00 (0/17)	0.64
8	0.35	0.674	0.646 [0.412, 0.860]	-0.062	0.33 (8/12)	0.75	0.47	0.849	0.803 [0.637, 0.934]	-0.105*	0.88 (2/17)	0.36
16	0.45	0.792	0.715 [0.496, 0.907]	+0.007	0.50 (6/12)	0.75	0.58	0.798	0.819 [0.657, 0.949]	-0.088*	0.94 (1/17)	0.36
32	0.55	0.833	0.785 [0.580, 0.950]	+0.076	0.50 (6/12)	0.75	0.67	0.790	0.840 [0.689, 0.958]	-0.067	0.94 (1/17)	0.43
64	0.64	0.819	0.792 [0.594, 0.957]	+0.083	0.50 (6/12)	0.75	0.76	0.739	0.853 [0.705, 0.965]	-0.055	0.94 (1/17)	0.50
128	0.73	0.646	0.778 [0.586, 0.950]	+0.069	0.50 (6/12)	0.75	0.84	0.676	0.866 [0.726, 0.971]	-0.042	1.00 (0/17)	0.50
256	0.83	0.743	0.778 [0.586, 0.950]	+0.069	0.50 (6/12)	0.75	0.91	0.655	0.870 [0.732, 0.974]	-0.038	1.00 (0/17)	0.50

Var.: cumulative explained variance on development cells. * $p < 0.05$ unadjusted; † $p < 0.05$ after Holm correction.

TABLE S5
ATTENTION MIL TRAINED FOR A FIXED 40 EPOCHS OR WITH THE NUMBER OF EPOCHS CHOSEN BY NESTED 3-FOLD CV INSIDE EACH TRAINING FOLD (LOWEST MEAN VALIDATION LOSS). EXTERNAL AUROC, RECALL, F1 AND CALIBRATION ERROR (ECE); SD: SPREAD OF DEVELOPMENT CV AUROC ACROSS SEEDS.

		External AUROC		F1		ECE		Chosen epochs
Dataset	Features	Fixed	Nested	Fixed	Nested	Fixed	Nested	median (range)
APL-AML	Engineered	0.812	0.819	0.67	0.67	0.26	0.26	18 (3–37)
	EfficientNet-B0	0.674	0.667	0.55	0.57	0.36	0.36	5 (3–7)
	Fusion	0.681	0.688	0.55	0.55	0.33	0.35	5 (2–8)
C-NMC	Engineered	0.912	0.895	0.81	0.81	0.25	0.27	33 (15–40)
	EfficientNet-B0	0.853	0.798	0.81	0.74	0.20	0.22	4 (1–36)
	Fusion	0.853	0.811	0.85	0.76	0.21	0.17	4 (1–40)

TABLE S6
PATIENT-LEVEL APL DETECTION (T3). CV: DISCOVERY, MEAN ± SD OVER THREE SEEDS. EXTERNAL: VALIDATION COHORT (12 APL, 12 AML) AT THE OPERATING POINT FIXED ON DISCOVERY (TARGET SENSITIVITY 0.95); THE LAST THREE COLUMNS USE A YOUDEN THRESHOLD FOR REFERENCE.

			External, sensitivity-first operating point						External, Youden threshold		
Features	Aggregation	CV AUROC	AUROC [95% CI]	Recall (missed)	Spec.	F1	NPV	ECE	Recall (missed)	Spec.	Acc.
Engineered	Pooled + LR	0.867±0.059	0.757 [0.555, 0.938]	1.00 (0/12)	0.17	0.71	1.00	0.294	0.42 (7/12)	0.92	0.67
EfficientNet-B0	Pooled + LR	0.909±0.018	0.722 [0.493, 0.914]	0.50 (6/12)	0.67	0.55	0.57	0.338	0.50 (6/12)	0.67	0.58
Fusion	Pooled + LR	0.910±0.018	0.708 [0.486, 0.909]	0.50 (6/12)	0.67	0.55	0.57	0.339	0.50 (6/12)	0.67	0.58
Engineered	Attention MIL	0.827±0.003	**0.812 [0.614, 0.963]**	1.00 (0/12)	0.00	0.67	0.00	0.264	0.50 (6/12)	0.75	0.62
EfficientNet-B0	Attention MIL	0.885±0.021	0.674 [0.444, 0.879]	0.50 (6/12)	0.67	0.55	0.57	0.362	0.50 (6/12)	0.75	0.62
Fusion	Attention MIL	0.886±0.039	0.681 [0.445, 0.879]	0.50 (6/12)	0.67	0.55	0.57	0.325	0.50 (6/12)	0.75	0.62

TABLE S7
SECONDARY ANALYSIS WITH MORE STATISTICAL POWER: REPEATED (3×) 5-FOLD PATIENT-LEVEL CV OVER ALL 106 APL-AML PATIENTS. THIS IS AN INTERNAL ANALYSIS, NOT AN EXTERNAL VALIDATION. Δ: AUROC DIFFERENCE VS. THE UNREDUCED EFFICIENTNET-B0 EMBEDDING (PAIRED PATIENT BOOTSTRAP ON REPEAT-AVERAGED SCORES). RECALL, SPECIFICITY AND F1 AT A THRESHOLD CHOSEN FOR SENSITIVITY ≥ 0.95 INSIDE EACH TRAINING FOLD.

Model	AUROC [95% CI]	Δ vs. no PCA	Recall (missed/34)	Spec.	F1
Promyelocyte fraction (biomarker)	0.677 [0.572, 0.784]	-0.275*†	1.00 (0.0)	0.00	0.49
Engineered, pooled LR	0.893 [0.823, 0.949]	-0.059*	0.94 (2.0)	0.42	0.59
EfficientNet-B0, no PCA	0.953 [0.898, 0.989]	–	0.95 (1.7)	0.53	0.64
EfficientNet-B0 + PCA-16	0.878 [0.802, 0.941]	-0.074*†	0.98 (0.7)	0.35	0.58
EfficientNet-B0 + PCA-32	0.947 [0.897, 0.985]	-0.005	0.96 (1.3)	0.60	0.69
EfficientNet-B0 + PCA-64	0.942 [0.892, 0.979]	-0.011	0.95 (1.7)	0.63	0.70
Engineered + PCA-32 fusion	0.922 [0.853, 0.973]	-0.030	0.95 (1.7)	0.58	0.67
Engineered, attention MIL	0.867 [0.794, 0.931]	-0.086*	0.89 (3.7)	0.52	0.62
EfficientNet-B0, no PCA‡	0.929 [0.870, 0.974]	–	0.96 (1.3)	0.48	0.63
EfficientNet-B0 + PCA-16‡	0.872 [0.791, 0.939]	-0.057*	0.96 (1.3)	0.31	0.56
EfficientNet-B0 + PCA-32‡	0.929 [0.866, 0.976]	+0.000	0.96 (1.3)	0.46	0.62
EfficientNet-B0 + PCA-64‡	0.935 [0.876, 0.979]	+0.006	0.97 (1.0)	0.49	0.63

* $p < 0.05$ unadjusted; † after Holm correction. ‡ PCA scores at natural scale; Δ vs. no PCA with the same scaling.

TABLE S8

CAN A NEW LABORATORY RESTORE THE OPERATING POINT WITH A FEW LOCAL CASES? k CONFIRMED LOCAL PATIENTS ($k/2$ APL, $k/2$ AML) ARE DRAWN FROM THE VALIDATION COHORT AND THE REMAINING $24 - k$ PATIENTS ARE SCORED (2,000 RANDOM DRAWS). R0: DEVELOPMENT THRESHOLD. R2: DEVELOPMENT THRESHOLD LOWERED, IF NEEDED, TO JUST BELOW THE LOWEST SCORE OF THE LOCAL APL CASES. P(MISS): PROBABILITY THAT AT LEAST ONE APL PATIENT IN THE TEST SET IS MISSED.

	R0, $k = 0$				R2, $k = 4$				R2, $k = 8$			
Model	Recall	Spec.	F1	P(miss)	Recall	Spec.	F1	P(miss)	Recall	Spec.	F1	P(miss)
EfficientNet-B0, no PCA	0.50	0.67	0.55	1.00	0.69	0.56	0.63	0.84	0.80	0.49	0.68	0.68
EfficientNet-B0 + PCA-64 (pre-decl.)	0.50	0.92	0.63	1.00	0.70	0.66	0.68	0.82	0.80	0.44	0.68	0.66
EfficientNet-B0 + PCA-16	0.92	0.67	0.81	1.00	0.92	0.62	0.80	0.83	0.92	0.58	0.79	0.67
Engineered, attention MIL	1.00	0.00	0.67	0.00	1.00	0.00	0.67	0.00	1.00	0.00	0.67	0.00
Fine-tuned EfficientNet-B0	0.83	0.67	0.77	1.00	0.85	0.63	0.76	0.84	0.86	0.59	0.76	0.67

TABLE S9

FEATURE-GROUP ABLATION ON THE EXTERNAL COHORT (ENGINEERED FEATURES, LR). Δ: CHANGE IN AUROC RELATIVE TO ALL 93 FEATURES, PAIRED PATIENT BOOTSTRAP.

	T2 Blast detection		T1 Diagnosis on cells	
Setting (n)	AUROC	Δ [95% CI]	AUROC	Δ [95% CI]
All features (93)	0.950	–	0.671	–
Without one group				
Nuclear shape (84)	0.949	-0.001 [-0.003, +0.001]	0.681	+0.010 [-0.004, +0.024]
Basophilic region (83)	0.945	-0.005 [-0.009, -0.003]*	0.665	-0.005 [-0.025, +0.010]
Stain and colour (67)	0.916	-0.034 [-0.060, -0.015]*	0.548	-0.123 [-0.272, +0.046]
Chromatin texture (77)	0.939	-0.012 [-0.023, -0.004]*	0.675	+0.004 [-0.007, +0.016]
Local binary patterns (65)	0.947	-0.004 [-0.010, +0.002]	0.670	-0.001 [-0.019, +0.020]
Nucleoli and granules (89)	0.951	+0.001 [-0.002, +0.004]	0.671	+0.000 [-0.006, +0.006]
One group only				
Nuclear shape (9)	0.810	-0.140 [-0.212, -0.082]*	0.549	-0.121 [-0.283, +0.062]
Basophilic region (10)	0.823	-0.127 [-0.198, -0.069]*	0.579	-0.092 [-0.242, +0.073]
Stain and colour (26)	0.912	-0.038 [-0.078, -0.013]*	0.682	+0.011 [-0.032, +0.041]
Chromatin texture (16)	0.802	-0.148 [-0.221, -0.086]*	0.526	-0.145 [-0.331, +0.067]
Local binary patterns (28)	0.793	-0.157 [-0.225, -0.098]*	0.581	-0.090 [-0.241, +0.081]
Nucleoli and granules (4)	0.860	-0.090 [-0.142, -0.054]*	0.500	-0.171 [-0.365, +0.057]

* $p < 0.05$.